

BEN KROTONSKY

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EDUCATION

Northern Arizona University (NAU)

Bachelor of Science, Mechanical Engineering, Minor in German | GPA: 4.0 / 4.0

Flagstaff, Arizona

Expected May 2027

WORK EXPERIENCE

Dynamic and Active Systems Lab (DASL) — NAU

Undergraduate Research Assistant

September 2023 – Present

Flagstaff, Arizona

- Engineered a statistical temperature variance model to distinguish anomalous signatures from background population noise.
- Conducted a soil moisture analysis at active leak sites to determine leak characteristics for POI verification.
- Created an automated POI identification algorithm achieving 40% true positive rate and 0.2% false positive rate without human interference across 10 verified leak sites.
- Designed a MATLAB GUI to streamline the thermal analysis interface and implemented parallel processing and workflow optimization, reducing total processing time by 70.4% versus the original workflow.
- Developing a controlled leak experiment to simulate pipe leak behavior in Phoenix by creating a scaled, two-dimensional soil/pipe cross-section to build high-accuracy models for soil moisture and define accurate leak behavior over time.
- Built orthomosaic stitching workflow in Agisoft Metashape aerial thermal analysis of water pipelines.
- Authored several standard operating procedure documents for drone operation and mission planning in DASL.
- Executed 20+ drone test missions using a DJI Mavic 3T across eight sites to acquire thermal imagery of known leaks and establish POI detection criteria.

match — Leibniz University Hannover (LUH)

DAAD RISE Engineering Research Intern

November 2024 – August 2026

Hannover, Germany

- Developed a geometry-based algorithm to quantify stable pose probabilities for arbitrary workpieces using Centroid Solid Angle (CSA) and Critical Solid Angle (CRSA) analyses.
- Adapted CSA/CRSA analyses from a normal plane to a non-normal bi-planar condition with multiple tilt angle considerations.
- Defined new conditions for instability, accounting for a part's resultant forces and motion after planar contact.
- Tested algorithm across 5 unique parts with 12 unique angle conditions using a custom-built PyBullet Monte-Carlo simulation, achieving a mean absolute error of 5.095% against simulation with a R^2 value of 0.794.
- Presented findings at the DAAD RISE Conference in Heidelberg, Germany.
- Formulated a reorientation algorithm to determine all possible paths between a part's stable poses, accounting for all reorientation axes and physically unstable poses, and identifying the most efficient reorientation order.
- Validated algorithm across 5 unique parts, achieving 100% accuracy for each part's pose reorientation network.

ENGINEERING LEADERSHIP

Society of Automotive Engineers (SAE)

Chapter Secretary

August 2025 – May 2026

Flagstaff, Arizona

- Designed and led a workshop on applied statistics and MATLAB fundamentals for ~10 members, translating research-level statistical methods into an accessible introductory session.

TECHNICAL SKILLS

Engineering Software: MATLAB, Python, SolidWorks, Autodesk Inventor, Agisoft Metashape, WebODM

Data & Analysis: Excel, MATLAB (numerical methods, nodal analysis, statistical modeling), PyBullet, Machine Learning

Field & Hardware: Drone Operation (DJI Mavic 3T, RTK), Thermal IR Imaging, Universal Robots

Methods: Finite Difference Analysis, Aerodynamic Analysis, Probabilistic Modeling, Algorithm Development, GUI Design

Languages: English (Native), German (Professional Working Proficiency, B2)